GGIB Part 4 - Human Baseline Design

Authors: Clarence Downs, Caroline Brooks
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Competition: Kaggle - Measuring Progress Toward AGI: Cognitive Abilities
Track: Metacognition
Normative standard: Ideal epistemic agent (locked in Part 1)

Why the Human Baseline Matters

From GGIB-Part1-Critical-Analysis, Section 5.6:

> Humans are bad at metacognition. Dunning-Kruger effects, overconfidence bias, anchoring, sunk-cost reasoning, and motivated reasoning are all well-documented metacognitive failures in humans. If the benchmark uses human baselines, it must decide whether it is measuring "does the AI behave like a human?" or "does the AI behave like an idealized epistemic agent?"

The decision is locked: the normative standard is the **ideal epistemic agent**. The human baseline is **descriptive, not normative**. It characterizes how humans actually perform on these tasks, providing a reference distribution against which AI performance can be contextualized - not a target to match.

This distinction is load-bearing. If the human baseline becomes the normative target, the benchmark rewards AI systems that replicate human metacognitive

biases (overconfidence, anchoring, social desirability). That is the wrong goal.

From Metacognition-Benchmark-Critique:

> Human epistemic expression varies enormously by culture, education, personality, domain expertise, and communication style. The prompt asks the model to address "how to normalize across different communication styles" but treats this as a sub-bullet rather than a central design challenge.

This part treats it as a central design challenge.

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1. Participant Selection

1.1 Sample Definition

Three participant groups, aligned with the baseline tiers in AGI-Benchmark-Blueprint:

Group	Label	Definition	Purpose
A	General Adult	Adults 18-65, no domain-specific expertise required, basic literacy in the task language	Establishes floor-level human metacognitive behavior
B	Educated Adult	Bachelor's degree or equivalent, comfortable with structured reasoning tasks	Primary comparison group for AI systems
C	Tool-Assisted Adult	Same as Group B, but permitted to use calculator, search engine, notes, and documentation during the task	Controls for the fact that AI systems have tool access; comparing tool-using AI to unassisted humans distorts conclusions

1.2 Inclusion Criteria

All groups:

- Age 18-65
- Fluent in the task administration language (English for V1)
- No diagnosed cognitive impairments affecting reasoning or reading comprehension
- Able to complete a 60-90 minute task session
- No prior exposure to the specific benchmark tasks

Group B and C additionally:

- Completed undergraduate degree or equivalent professional experience (4+ years in a knowledge-work field)
- Self-reported comfort with structured analytical tasks

1.3 Exclusion Criteria

- Professional AI researchers or ML engineers (to prevent pattern recognition of benchmark structure)
- Professional psychometricians or assessment designers (same concern)
- Anyone who has seen the benchmark tasks in any form

1.4 Minimum Sample Size

Per group: N = 50 minimum. Target: N = 80.

Rationale: The primary analysis is distributional, not point-estimate. With N = 50 per group, we can estimate the 25th, 50th, and 75th percentiles of each scoring dimension with reasonable precision. With N = 80, we can also estimate the 10th and 90th percentiles. Smaller samples produce distributions too noisy to serve as meaningful reference points.

Power analysis basis: For detecting a medium effect size (Cohen's d = 0.5) between Group A and Group B on any scoring dimension at alpha = 0.05 with power = 0.80, the minimum per-group N is approximately 64. N = 80 provides a buffer for attrition and data quality exclusions.

Alternative rejected: N = 20 per group. Rejected because distributional estimation with N = 20 is unstable, and the benchmark's comparison method is distributional. Point-estimate comparison (mean vs. mean) would tolerate smaller samples but is inappropriate for this design.

1.5 Recruitment

- Group A: Recruited via online platforms (Prolific, MTurk) with demographic screening. Payment at or above platform median for task duration.
- Group B: Recruited via university alumni networks and professional organizations. Incentive: payment plus optional summary report of results.
- Group C: Same recruitment as Group B. Participants randomly assigned to Group B or C to control for selection effects.

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2. Task Administration

2.1 Modality

Text-based, self-paced, browser-administered.

Participants complete tasks through a web interface that presents each task prompt and collects structured responses in the same format required of AI systems. The interface enforces the required output fields (ANSWERABLE, CONFIDENCE, REASONING, etc.) through form elements.

Alternative rejected: Think-aloud protocol with audio recording. Rejected for V1 because think-aloud adds substantial coding burden, introduces social desirability effects (participants perform for the recorder), and is incompatible with the scale needed for N = 80 per group. Think-aloud could be added as a supplementary study for a subset of participants in a later version.

2.2 Task Presentation

- Tasks are presented in randomized order within each family to prevent sequence effects.
- Task families are presented in blocks (all EGD tasks together, then all CDSP tasks, etc.) with short breaks between blocks, to reduce cognitive switching costs.
- Block order is counterbalanced across participants using a Latin square design.
- Each participant completes a subset of the full task inventory (not all 100+ instances). Target: 30 tasks per participant (6 per family), selected via stratified random sampling to ensure coverage of all evidence conditions within each family.

2.3 Timing

- No per-task time limit. Self-paced.
- Total session time: estimated 60-90 minutes including breaks.
- Session can be paused and resumed within a 48-hour window.

Rationale for no time limit: Time pressure introduces a confound. Under time pressure, humans hedge more (or less, depending on personality), and the effect of time pressure on epistemic behavior is a separate variable from metacognitive capacity. Since the AI systems being evaluated are not time-pressured on individual tasks, adding time pressure to the human baseline would distort the comparison.

Alternative rejected: 3-minute per-task limit. Rejected because it confounds processing speed with metacognitive quality. Some tasks require careful reading of multiple evidence sources. Fast participants would be advantaged in ways unrelated to metacognition.

2.4 Instructions

Participants receive:

1. **General framing:** "This study examines how people evaluate evidence and express confidence in their assessments. There are no trick questions, but

some tasks have more evidence than others. Your job is to assess the evidence honestly, not to get the 'right' answer."

2. **Format training:** A walkthrough of the structured response format (ANSWERABLE, CONFIDENCE, REASONING, MISSING, etc.) with two practice tasks that are not scored. Practice tasks include one sufficient-evidence and one insufficient-evidence example with feedback showing the expected response.
3. **Confidence scale anchoring:** "CONFIDENCE means how confident you are that your assessment is appropriate given the evidence. 0 = no confidence at all; 100 = completely certain. There is no penalty for low confidence when evidence is weak. There is also no penalty for high confidence when evidence is strong."
4. **No deception:** Participants are told that some tasks will have incomplete or contradictory evidence, and that recognizing this is a valid and valued response.

2.5 Revision Capability

For Task Family 4 (SEI), the multi-turn structure is preserved:

- Turn 1 response is submitted and locked.
- New evidence is presented.
- Turn 2 response is collected.
- Participants cannot go back and edit Turn 1 after seeing Turn 2 evidence.

For all other families, participants can revise their answers before final submission of each task, but not after submission.

2.6 Tool Access (Group C Only)

Group C participants are provided:

- A web browser tab for search (no restrictions on search queries)
- A calculator application
- A notepad for scratch work

They are instructed: "You may use these tools however you like. Use them as you normally would when trying to assess a complex question."

Tool usage is logged (search queries, calculator entries, time spent with tools) but not scored. The purpose is to characterize how tool access changes human epistemic behavior, not to evaluate tool usage quality.

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3. Coding Human Epistemic Behavior

3.1 Structured Fields (Automated Scoring)

Because the human participants use the same structured output format as the AI systems, the primary scoring dimensions from GGIB Part 3 - Formal Scoring Framework can be applied directly:

- **EJA (Epistemic Judgment Accuracy):** Categorical match, same computation as for AI.
- **CA (Confidence Appropriateness):** Same range-deviation scoring as for AI.
- **MEI (Missing Element Identification):** Same F1 set-comparison as for AI.
- **RA (Revision Appropriateness):** Same cross-turn analysis as for AI.

This is a major design advantage of using structured output: the same scoring pipeline runs on both human and AI responses without modification.

3.2 Reasoning Field (Requires Calibrated Scoring)

The RS (Reasoning Specificity) dimension uses a dual-path matcher architecture (see GGIB Part 3 - Formal Scoring Framework v0.2). Path A checks whether the reasoning references a key evidence element. Path B checks whether the element is functionally utilized in the reasoning chain (transformed, compared, linked to the epistemic constraint), not merely mentioned. For human

responses, the same dual-path matcher is applied, but human writing varies more in style, length, and directness.

Calibration requirements: Both matcher paths must be validated against human responses specifically.

- **Path A (reference presence):** A pilot set of 50 human reasoning responses per family is annotated by 2 independent raters for each key evidence element (present/absent in the reasoning). Inter-rater reliability target: Cohen's kappa >= 0.75.
- **Path B (reference utilization):** The same pilot set is annotated for whether each referenced element is functionally used or merely mentioned. The contrastive grader is calibrated against these annotations. Target: 85% agreement with human gold labels on a set of 50 contrastive pairs.

3.3 Uncertainty Expression Coding

Human uncertainty expression is more varied than AI output. Humans use:

- Explicit hedging: "I'm not sure," "probably," "it depends"
- Conditional language: "If X is true, then Y"
- Question-asking: "I'd want to know whether..."
- Emotional hedging: "I feel like maybe..."
- Silence/omission: leaving fields blank or writing very short responses

The structured format constrains most of this variation (participants must provide a CONFIDENCE number and a categorical ANSWERABLE judgment), but the REASONING field still captures natural variation.

Coding protocol: No additional coding beyond the automated scoring pipeline is required for primary metrics. The structured format handles the conversion from natural human expression to scored output.

For supplementary analysis, a subset of 100 REASONING responses (stratified by evidence condition and participant group) are coded by 2 raters on a 4-point scale (updated for v0.2 dual-path matcher):

Code	Label	Definition
0	No evidence engagement	Reasoning contains no reference to specific evidence elements
1	Reference without utilization	Reasoning mentions evidence elements but does not use them to drive the epistemic judgment
2	Partial utilization	Reasoning functionally uses some key evidence elements but misses others
3	Full utilization	Reasoning functionally uses the key evidence elements to support or constrain the epistemic judgment

This supplementary coding validates both paths of the automated RS scoring against human judgment. The 0/1 boundary validates Path A (reference presence). The 1/2 and 2/3 boundaries validate Path B (reference utilization). It is not required for the primary baseline computation.

3.4 Handling Missing or Invalid Responses

- If a participant leaves a structured field blank, that task instance is excluded from that dimension's scoring but not from other dimensions.
- If a participant provides a CONFIDENCE value outside [0, 100], it is capped at the nearest boundary.
- If a participant writes "I don't know" or equivalent in the ANSWER field but marks ANSWERABLE: yes, the behavioral signals are contradictory. Both signals are recorded; the ANSWERABLE field takes precedence for EJA scoring.

4. Normalizing Across Communication Style

4.1 The Problem

Humans vary dramatically in how they express the same epistemic state:

- **Verbose vs. concise:** One person writes three paragraphs of reasoning; another writes one sentence. Both may reference the same evidence elements.
- **Direct vs. hedged:** Some people state conclusions bluntly ("There's not enough data"); others embed conclusions in qualifiers ("It seems like there might potentially be some gaps in the information provided").
- **Confident vs. cautious personality:** Some people habitually express high confidence; others habitually express low confidence regardless of evidence quality.
- **Cultural variation:** Directness norms differ across cultures. What reads as "appropriate confidence" in one cultural context reads as "arrogance" in another.

4.2 Design Mitigation (Built Into the Protocol)

The structured output format is the primary mitigation. By requiring participants to provide explicit ANSWERABLE, CONFIDENCE, REASONING, and MISSING fields, the protocol converts natural variation into structured data that can be scored uniformly.

This does not eliminate style variation in the REASONING field, but it constrains the scoring-relevant output to structured fields that are less sensitive to communication style.

4.3 Statistical Normalization

Within-participant z-scoring for confidence: Each participant's CONFIDENCE values are z-scored within their own distribution before cross-participant comparison. This removes the "habitually high/low confidence" confound. A participant who always gives CONFIDENCE: 70-90 and another who always gives CONFIDENCE: 30-50 can both show appropriate evidence-sensitivity if their within-participant variation tracks evidence quality.

```
For participant p:
  z_confidence_p_i = (confidence_p_i - mean(confidence_p)) /
std(confidence_p)
```

The z-scored confidence is used for distributional comparison. The raw confidence is used for CA scoring (because CA scores against absolute ranges defined by the ideal epistemic agent standard, not relative to individual baselines).

Why both raw and z-scored: Raw scoring answers "Did this participant behave like an ideal epistemic agent?" Z-scored analysis answers "Did this participant's confidence track evidence quality, regardless of their personal calibration baseline?" Both are informative.

4.4 What Cannot Be Normalized

Genuine differences in metacognitive ability cannot and should not be normalized away. If some participants consistently fail to detect contradictions while others catch them, that is signal, not noise. The normalization applies only to communication-style variation (how they express their judgments), not to judgment quality itself.

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5. Known Human Biases - Mitigation Strategies

5.1 Overconfidence Bias

The bias: Humans systematically overestimate the accuracy of their judgments, particularly in domains where they have moderate knowledge. This is one of the most robust findings in judgment and decision-making research.

Expected effect on baseline: Human confidence scores will skew high relative to the ideal epistemic agent standard, particularly on insufficient-evidence and

contradictory-evidence tasks. Humans will express CONFIDENCE: 60-80 where the ideal epistemic agent would express CONFIDENCE: 10-30.

Mitigation: The confidence anchoring instructions explicitly state that low confidence is not penalized. This reduces (but does not eliminate) the social pressure to appear confident. The within-participant z-scoring further separates evidence-sensitivity from absolute calibration level.

Design note: Overconfidence in the human baseline is not a problem to solve. It is a datum to capture. The benchmark uses the ideal epistemic agent as the normative standard, not human performance. Documenting human overconfidence makes the baseline more useful, not less.

5.2 Anchoring

The bias: Initial information disproportionately influences subsequent judgments. In the context of this benchmark, participants may anchor on the first piece of evidence they read and insufficiently adjust when they encounter contradicting or qualifying information.

Expected effect on baseline: In Task Family 4 (SEI), participants will under-revise after disconfirming evidence (anchoring to their Turn 1 response). In Task Family 2 (CDSP), participants may anchor on the first source presented and discount contradicting sources that appear later.

Mitigation:

- Evidence source order is randomized across task instances within CDSP to prevent systematic anchoring on the first-listed source.
- In SEI tasks, the multi-turn structure inherently tests anchoring resistance. No mitigation is needed because anchoring behavior is part of what the baseline should capture.

5.3 Social Desirability

The bias: Participants adjust their behavior to appear more competent, rational, or careful than they would naturally be. In this benchmark, social desirability could manifest as over-hedging (expressing more uncertainty than genuinely

felt) or over-thoroughness (listing more missing evidence elements than they would naturally consider).

Expected effect on baseline: Human RS and MEI scores may be inflated in the study setting relative to how the same individuals would behave in natural decision-making contexts. This is a fundamental limitation of any controlled study.

Mitigation:

- Instructions frame honesty as the goal, not thoroughness: "Your job is to assess the evidence honestly, not to get the 'right' answer."
- The study does not provide feedback during the task session (no "you got this one right" signals that could train participants to adopt a strategy).
- Anonymity: Participants are identified by code only. Responses are not linked to identifiable information in any reporting.

Residual risk: Social desirability cannot be fully eliminated. The baseline should be interpreted with this bias acknowledged. Supplementary analysis could compare early-session vs. late-session responses to check whether participants become more strategic (more hedging, more missing-element listing) as they learn the task structure.

5.4 Fatigue and Task Learning Effects

The bias: Performance changes over the course of a session due to fatigue (degradation) or practice (improvement). Both effects are expected.

Expected effect on baseline: Earlier tasks may show lower RS and MEI scores (participants haven't learned the format yet). Later tasks may show lower CA precision (fatigue reduces careful thinking).

Mitigation:

- Counterbalanced block order (Latin square) ensures that no task family is systematically positioned early or late for all participants.
- Breaks between blocks reduce fatigue accumulation.

- First 2 tasks in each block can be analyzed separately as "warm-up" items and optionally excluded from the primary analysis if they show systematic differences from later items.

5.5 Dunning-Kruger Effects

The bias: Individuals with lower competence in a domain tend to overestimate their performance, while highly competent individuals tend to slightly underestimate theirs.

Expected effect on baseline: Group A (general adult) may show higher CONFIDENCE values on difficult tasks than Group B (educated adult), not because they perform better, but because they are less aware of task difficulty. Group B may show slightly lower CONFIDENCE than their actual performance warrants.

Mitigation: This is captured in the distributional comparison, not normalized away. The three-group structure (A, B, C) allows the analysis to characterize how metacognitive calibration varies with baseline competence. This is valuable data, not noise.

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6. Comparison Method: AI vs. Human Baseline

6.1 Distributional Comparison (Primary Method)

The comparison is not "Is the AI better or worse than humans?" It is "Where does the AI's epistemic behavior fall within the human distribution?"

For each scoring dimension (EJA, RS, CA, RA, MEI, ECS):

1. Compute the human distribution from Group B (primary comparison group).
2. Compute percentile values: 10th, 25th, 50th (median), 75th, 90th.
3. Place the AI system's score within this distribution.

Output format:

```
Dimension: EJA
Human baseline (Group B, N=80):
  10th percentile: 0.52
  25th percentile: 0.61
  Median: 0.72
  75th percentile: 0.81
  90th percentile: 0.88

AI system score: 0.84
Percentile rank: 78th (above median, below 90th)
```

This is more informative than a single "beats human / does not beat human" comparison because it shows where the AI falls in the human distribution and how much variation exists among humans.

6.2 Profile Similarity (Secondary Method)

Across the five task families, the AI system and the human baseline each produce a profile of scores. Profile similarity measures whether the AI system's pattern of strengths and weaknesses resembles the human pattern.

Method: Compute the Spearman rank correlation between the AI system's per-family ECS scores and the median human ECS scores per family.

- High correlation (rho > 0.7): The AI system's metacognitive profile is shaped like the human profile (strong where humans are strong, weak where humans are weak).
- Low correlation (rho < 0.3): The AI system's metacognitive profile is differently shaped from the human profile.

This captures jaggedness. An AI system that scores well overall but has a very different profile shape than humans is exhibiting a qualitatively different kind of metacognitive behavior.

6.3 Calibration Comparison (Tertiary Method)

For Dimension 3 (CA), compare the AI system's confidence-evidence correlation against the human distribution of confidence-evidence correlations.

Method:

1. For each human participant, compute Spearman's rho between their stated CONFIDENCE values and the labeled evidence quality levels across all tasks.
2. This produces a distribution of human calibration coefficients.
3. Compute the same coefficient for the AI system.
4. Place the AI system's calibration coefficient within the human distribution.

This specifically tests whether the AI system is better or worse calibrated than typical humans - relevant because humans are known to be poorly calibrated, and an AI system that is *better* calibrated than humans is potentially more valuable than one that merely matches human calibration.

6.4 What the Comparison Does Not Do

The comparison does not:

- Declare the AI system "has metacognition" or "lacks metacognition" based on human comparison
- Use the human median as a pass/fail threshold
- Average across scoring dimensions into a single comparison number
- Assume that matching human performance is the goal (it is not; the ideal epistemic agent standard governs scoring)

The human baseline is a reference frame, not a success criterion.

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7. Reliability Requirements

7.1 Inter-Rater Reliability for Supplementary Coding

The supplementary 4-point coding of REASONING responses (Section 3.3) requires:

- 2 independent raters per response
- Training set: 20 responses coded jointly with discussion to establish shared standards
- Test set: 100 responses coded independently
- Minimum threshold: Cohen's kappa >= 0.75 on the test set
- If threshold is not met: retrain, revise coding guide, re-test with new responses

7.2 Inter-Rater Reliability for Evidence-Element Matching Calibration

The dual-path matcher calibration (Section 3.2, aligned with GGIB Part 3 - Formal Scoring Framework v0.2) requires:

Path A (reference presence):

- 2 independent raters annotating 50 human reasoning responses per family for presence/absence of each key evidence element
- Minimum threshold: Cohen's kappa >= 0.75 per evidence element
- Elements where kappa < 0.75: revise element definition, re-annotate, or exclude from RS scoring

Path B (reference utilization):

- Same raters annotating whether each referenced element is functionally used or merely mentioned
- Contrastive grader validated against 50 human-annotated contrastive pairs
- Minimum threshold: 85% agreement with human gold labels
- If threshold is not met: revise grading prompt or switch grading models

7.3 Test-Retest Reliability

A subset of 20 participants from Group B are invited to repeat the session 2-4 weeks later with a different random subset of tasks from the same families.

This provides:

- Test-retest correlation for each scoring dimension
- Evidence that the human baseline is stable, not an artifact of a single session

Target: Pearson r >= 0.60 for ECS across sessions. If test-retest reliability is below this threshold, the task design may need revision (tasks may be too variable in difficulty or too sensitive to momentary attention).

7.4 Internal Consistency

Within each task family, compute Cronbach's alpha across the 6 task instances each participant completes.

Target: alpha >= 0.70 per family. This indicates that the tasks within a family are measuring the same construct. If alpha is low, the tasks within a family may be too heterogeneous.

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8. Practical Execution Plan

8.1 Timeline (Within the 4-Week Build Window)

Week	Activity
1	Build web-based task administration interface. Finalize task subset selection (30 tasks per participant). Write participant instructions and consent form.
2	Recruit 10 pilot participants (Group B only). Run pilot. Check for task comprehension issues, interface bugs, timing estimates.
3	Analyze pilot data. Calibrate dual-path matcher (Path A reference thresholds + Path B contrastive grader) against pilot REASONING responses. Revise tasks or interface as needed. Begin full recruitment.

Week	Activity
4	Complete data collection for Group B (N = 80). Run automated scoring pipeline. Compute distributional baselines.

Scoping note: If the 4-week timeline does not permit full execution of all three groups, prioritize Group B (the primary comparison group). Group A and Group C can be collected in a subsequent phase without invalidating the initial baseline.

8.2 Cost Estimate

- Pilot (10 participants, ~90 min each): ~$300 (at $20/hr)
- Group A (50-80 participants, ~90 min each): ~$1,500-2,400
- Group B (50-80 participants, ~90 min each): ~$1,500-2,400
- Group C (50-80 participants, ~90 min each): ~$1,500-2,400
- Web platform hosting and development: ~$200-500
- Supplementary coding labor (2 raters, ~20 hrs each): ~$800

Total estimated range: $5,800-8,700 for the full three-group design. $2,000-3,200 for Group B only (minimum viable baseline).

8.3 IRB Considerations

This study involves human participants completing cognitive tasks. Depending on institutional affiliation:

- University-affiliated researchers: IRB approval required before recruitment.
- Independent researchers: Consider using a commercial IRB service (e.g., WCG IRB) or ensure the study qualifies as exempt (minimal risk, anonymous data, no deception).
- The study involves no deception, no sensitive personal data collection, and minimal risk. It is likely to qualify for expedited or exempt review.

9. Limitations of the Human Baseline

These should be stated explicitly in any submission or publication:

1. **Ecological validity:** Participants are completing structured tasks in a study setting. This does not fully reflect how humans handle uncertainty in natural decision-making contexts, where stakes, time pressure, and social dynamics are different.
2. **Format constraint:** The structured output format (ANSWERABLE, CONFIDENCE, REASONING, MISSING) constrains natural expression. Some humans may express metacognitive awareness in ways the format does not capture (e.g., facial expressions, tone of voice, spontaneous questions).
3. **Sample bias:** Online recruitment skews toward younger, more educated, more tech-comfortable individuals. The baseline may not represent the full range of adult metacognitive behavior.
4. **Social desirability:** Study participation increases self-monitoring, potentially inflating metacognitive scores relative to natural behavior.
5. **Single session:** A one-session baseline does not capture within-person variation across contexts, fatigue levels, or emotional states. The test-retest subset partially addresses this but does not fully resolve it.
6. **Language constraint:** V1 is English-only. Metacognitive expression varies across languages and cultures.

Related Notes

- AGI-Benchmark-Blueprint
- GGIB-Part1-Critical-Analysis
- GGIB Part 2 - Metacognition Task Family Design
- GGIB Part 3 - Formal Scoring Framework

- Orchestrator_Epistemic_Capture_v2
- Agent_State_Framework_v3
- Metacognition-Benchmark-Critique
- Metacognition-Benchmark-Synthesis-Report

Tags

#AGI #benchmark #metacognition #human-baseline #evaluation-design #psychometrics #GGIB #Kaggle #competition #methodology